

THE STRUCTURE OF MATHEMATICAL REALITY

Phase V — Chapter 16

Category Theory: The Mathematics of Mathematics

Exercises

Self-Study Curriculum

1. In Chapter 7, you learned about group homomorphisms. In Chapter 8, about linear maps. In Chapter 9, about continuous functions. State precisely what all three have in common, then state how they differ. What is the categorical generalization?

2. The determinant is a function from matrices to numbers: $\det(AB) = \det(A)\det(B)$. In category-theoretic language, what is the determinant? (Hint: it is a map from one category to another that preserves a certain structure.)

3. Consider the “forgetful functor” from Grp (groups) to Set (sets): it takes a group and “forgets” its operation, leaving just the underlying set. What information is lost? What is preserved? Can you think of an analogue in your own work, a procedure that “forgets” structure?

4. In machine learning, a “feature map” takes raw data and maps it into a higher-dimensional space where it becomes linearly separable. In categorical terms, what is a feature map doing? What categories are involved?
